# INT 221 - MVC PROGRAMMING

## Unit 5

# CSRF Field

**Cross-Site Request Forgery** protection.

Prevents unauthorized form submissions.

**Laravel Implementation**

@csrf: -> Generates hidden input:

<input type="hidden" name="_token" value="...">

# Method Field

HTML forms support only:

GET, POST

Laravel allows:

PUT, PATCH, DELETE

**Syntax**

@method('PUT')

# Laravel Form Validation

Checking user input before processing.

**Example**

```
$request->validate([
   'name' => 'required',
]);
```

# Validation Rules

| Rule | Meaning |
|---|---|
| required | Field must be filled |
| email | Valid email |
| min:3 | Minimum Length |
| max:10 | Maximum Length |
| numeric | Must be number |

## Error Messages

**Default**

$errors->all()

## Custom Messages

```
$request->validate([
        'name' => 'required'
    ], [
        'name.required' => 'Name is mandatory.'
]);
```

## Custom Validation Rules

When default rules are not enough

Example: Name must start with capital letter

## Repopulating Forms (Old Input)

value="{{ old('name') }}"

Keeps user input after validation error

# MINI PROJECT

## Student Registration System (Advanced Form Handling), Includes:

CSRF, Method Field, Validation + Rules, Custom Rule, Error Messages, Repopulating Form and Flash Message

## PROJECT FLOW (IMPORTANT FOR EXAM)

Form → Submit → Validation
→ Error → Back with old input
→ Success → Store + Show Result

## STEP 1: Create Laravel Project

```
composer create-project laravel/laravel student-app
cd student-app
php artisan serve
```

Open: http://127.0.0.1:8000

**STEP 2: Create Controller->** php artisan make:controller StudentController

**STEP 3: Create Custom Validation Rule->** php artisan make:rule CapitalName

**app/Rules/CapitalName.php**

```
<?php
namespace App\Rules;
use Closure;
use Illuminate\Contracts\Validation\ValidationRule;

class CapitalName implements ValidationRule{
   public function validate(string $attribute, mixed $value, Closure $fail): void{
       if (!ctype_upper(substr($value, 0, 1))) {
           $fail('The :attribute must start with a capital letter.');
       }
   }
}
```

## STEP 4: Define Routes

routes/web.php

```
use Illuminate\Support\Facades\Route;
use App\Http\Controllers\StudentController;

// Show form
Route::get('/student', [StudentController::class, 'form'])->name('student.form');

// Submit form
Route::post('/student', [StudentController::class, 'submit'])->name('student.submit');

// Method field demo (PUT)
Route::put('/student/update', [StudentController::class, 'update'])->name('student.update');
```

# STEP 5: Controller Code

app/Http/Controllers/StudentController.php

```
<?php
namespace App\Http\Controllers;
use Illuminate\Http\Request;
use App\Rules\CapitalName;
class StudentController extends Controller{
        // Show Form
        public function form(){
            return view('student_form');
        }
              // Handle Form Submission
              public function submit(Request $request){
             // VALIDATION
            $validated = $request->validate([
                'name' => ['required', 'min:3', 'max:20', new CapitalName],
                'email' => 'required|email',
                'age' => 'required|numeric|min:18|max:60'
            ],
```

```
            [
            'name.required' => 'Name is required!',
            'name.min' => 'Name must be at least 3 characters',
            'email.required' => 'Email is required!',
            'email.email' => 'Enter valid email!',
            'age.min' => 'Age must be at least 18',
            'age.max' => 'Age must be less than 60'
        ]);
        // FLASH MESSAGE
        session()->flash('success', 'Student Registered Successfully!');
        // PASS DATA
        return view('student_result', ['data' => $validated]);
       }
    // Method Field Demo
    public function update()
    {
        return "PUT method called successfully!";
    }
}
```

# STEP 6: Create two views

php artisan make:view student_form
php artisan make:view student_result

# STEP 7: Create Form View->resources/views/student_form.blade.php

```
<!DOCTYPE html>
<html>
    <head>
        <title>Student Form </title>
        <!-- Bootstrap -->
        <link
      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"rel="stylesheet">
    </head>
    <body class="container mt-5">
        <h2>Student Registration </h2>
        <!-- SUCCESS MESSAGE -->
        @if(session('success'))
          <div class="alert alert-success">
            {{ session('success') }}
          </div>
        @endif
```

```
<form action="{{ route('student.submit') }}" method="POST">
    <!-- CSRF -->
    @csrf
        <!-- NAME -->
    <div class="mb-3">
        <label>Name:</label>
        <input type="text" name="name" class="form-control" value="{{ old('name') }}">
        @error('name')
            <small class="text-danger">{{ $message }}</small>
        @enderror
    </div>
    <!-- EMAIL -->
    <div class="mb-3">
        <label>Email:</label>
        <input type="email" name="email" class="form-control" value="{{ old('email') }}">
        @error('email')
            <small class="text-danger">{{ $message }}</small>
        @enderror
    </div>
```

```
        <!-- AGE -->
          <div class="mb-3">
              <label>Age:</label>
              <input type="number" name="age" class="form-control" value="{{ old('age') }}">
              @error('age')
                  <small class="text-danger">{{ $message }}</small>
              @enderror
          </div>
              <button class="btn btn-primary">Submit</button>
        </form><hr>
        <!-- METHOD FIELD DEMO -->
        <h4>PUT Method Demo</h4>
        <form action="{{ route('student.update') }}" method="POST">
                @csrf
                @method('PUT')
          <button class="btn btn-warning">Call PUT Route</button>
        </form>
    </body>
</html>
```

**STEP 8: Create Result View->** resources/views/student_result.blade.php

```
<!DOCTYPE html>
<html>
    <head>
        <title>Result</title>
    </head>
    <body>
        <h2>Student Details</h2>
        Name: {{ $data['name'] }} <br>
        Email: {{ $data['email'] }} <br>
        Age: {{ $data['age'] }} <br><br>
        <a href="{{ route('student.form') }}">Back</a>
    </body>
</html>
```

## STEP 9: Run Project

php artisan serve

Open: http://127.0.0.1:8000/student

# TEST CASES

**Case 1: Empty Form->** Errors appear + old values retained

**Case 2: Name not capital->** rabaab

Error: -> The name must start with a capital letter.

**Case 3: Correct Data**

Name: Rabaab
Email: test@gmail.com
Age: 22

Output: -> Student Registered Successfully!

The next page shows data

**Case 4: PUT Button**

Click button

Output:->PUT method called successfully!